EMBEDDING DIMENSION STRATA IN A SPACE OF COMPLETE SUBALGEBRAS OF POWER SERIES

KNOWLEDGE SENSE

ABSTRACT. Hamilton parametrized the complete $\mathbb{C}$-subalgebras of $\mathbb{C}[\![t]\!]$ with a prescribed value semigroup Γ by an affine variety $\mathcal{R}_\Gamma$. Hamilton asked whether the locus $\mathcal{R}_\Gamma^{(d)}$ of subalgebras generated by at most d elements is cut out by the nonvanishing of leading coefficients of fixed polynomials, and whether it has the form $(\mathbb{C}^*)^{g+1-d} \times \mathbb{C}^{N-g+d-1}$ when $\mathcal{R}_\Gamma \cong \mathbb{C}^N$. For $\Gamma = \langle 10, 11, 34, 46, 58\rangle$ we determine $\mathcal{R}_\Gamma$ and all its embedding dimension strata: there are global polynomial coordinates (q, A, B, C) identifying $\mathcal{R}_\Gamma$ with $\mathbb{A}^{54} \times \mathbb{A}^3$, in which the embedding dimension of a subalgebra equals $5 - \operatorname{rank} \left(\begin{smallmatrix} A & C \\ 0 & B \end{smallmatrix}\right)$. Consequently $\mathcal{R}_\Gamma^{(4)} \cong \mathbb{A}^{54} \times (\mathbb{A}^3 \setminus \{0\})$. This smooth open subset of $\mathcal{R}_\Gamma \cong \mathbb{A}^{57}$ is not affine, is not the nonvanishing locus of a single polynomial, and is not even homeomorphic to $\mathbb{C}^* \times \mathbb{C}^{56}$. Hence no fixed polynomial detects $\mathcal{R}_\Gamma^{(4)}$ through the leading coefficient of its reduction, as asked in Hamilton's Question 4.7, and $\mathcal{R}_\Gamma^{(4)}$ is not isomorphic to the space $\mathbb{C}^* \times \mathbb{C}^{56}$ proposed in Hamilton's Question 4.8.

1. INTRODUCTION

The analytic classification of unibranch curve singularities leads to the classification of complete $\mathbb{C}$-subalgebras of the power series ring $\mathbb{C}[\![t]\!]$. The semigroup of orders of the elements of such a subalgebra is a discrete invariant, and Zariski's moduli problem [Zar86] asks for a description of the singularities with a given semigroup up to analytic equivalence. A finer and more rigid object is the set

$$\mathcal{R}_\Gamma = \{\text{complete } \mathbb{C}\text{-subalgebras of } \mathbb{C}[\![t]\!] \text{ with semigroup } \Gamma\}$$

of *embedded* subalgebras. Ishii [Ish80] showed that the corresponding moduli functor is represented by an affine scheme; see the discussion in the introduction of [Ham20]. Hamilton [Ham20] gave an explicit algorithmic description: every element of $\mathcal{R}_\Gamma$ has a unique generating set in normal form, and $\mathcal{R}_\Gamma$ is the zero set of explicit polynomials in the coefficients of these normal forms [Ham20, Theorem 2.22]. Hamilton also explained that $\mathcal{R}_\Gamma$ parametrizes the ways of gluing a singular point with semigroup Γ onto a fixed pointed smooth curve, and that the Zariski moduli space is the quotient of $\mathcal{R}_\Gamma$ by $\operatorname{Aut}\mathbb{C}[\![t]\!]$ [Ham20, Sections 1 and 4.3]. In this paper, as in the two questions discussed below, no such quotient is taken: two subalgebras that differ by a change of the parameter t are different points of $\mathcal{R}_\Gamma$. This also distinguishes $\mathcal{R}_\Gamma$ from crimping spaces, which parametrize singularities of a fixed analytic type; see [Bat24, Section 1.6] and the references there.

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Let $\Gamma = \langle v_0, \dots, v_g \rangle$ with minimal generators $v_0 < \cdots < v_g$. The variables written $x_0, \dots, x_g$ in [Ham20] are denoted here by $\xi_0, \dots, \xi_g$, and normal-form generators by $\eta_0(t), \dots, \eta_g(t)$, because x_n will denote a coefficient of a generator. An element of $\mathcal{R}_\Gamma$ is generated by $g+1$ normal-form elements, but it may be generated by fewer elements [Ham20, Remark 2.5]. Hamilton denotes by $\mathcal{R}_\Gamma^{(n)}$ the subset of subalgebras which can be generated by n elements [Ham20, Definition 4.4]; this gives a filtration of $\mathcal{R}_\Gamma$ by embedding dimension. For semigroups with three generators the plane locus is described in [Ham20, Section 4.2], and for arbitrary semigroups the following two questions are posed. In the first one, $\operatorname{red}_{\langle v_0,\dots,v_{i-1}\rangle}$ is the reduction of [Ham20, Algorithm 2.15], which removes from an element of $\mathbb{C}[t]/(t^c)$ the powers lying in $\langle v_0, \dots, v_{i-1} \rangle$ by means of the first i normal-form generators, and $\varphi_R \colon \mathbb{C}[\xi_0, \dots, \xi_g] \to \mathbb{C}[t]/(t^c)$ sends ξ_i to the i-th normal-form generator $\eta_i(t)$.

Question 1.1 ([Ham20, Question 4.7])**.** Can we explicitly write down polynomials $f_d, \dots, f_g \in \mathbb{C}[\xi_0, \dots, \xi_g]$ such that $R = \mathbb{C}[\eta_0(t), \dots, \eta_g(t)]$ lies in $\mathcal{R}_\Gamma^{(d)}$ if and only if $\operatorname{ord} \operatorname{red}_{\langle v_0,\dots,v_{i-1}\rangle}(\varphi_R(f_i)) = v_i$ for all $i \in \{d, \dots, g\}$?

If so, the leading coefficients of these reduced elements would exhibit $\mathcal{R}_\Gamma^{(d)}$ as a principal open subvariety of $\mathcal{R}_\Gamma$, as Hamilton notes after the question.

Question 1.2 ([Ham20, Question 4.8])**.** If $\mathcal{R}_\Gamma$ is in bijection with an affine space $\mathbb{C}^N$, can we always identify $\mathcal{R}_\Gamma^{(d)}$ with the space $(\mathbb{C}^*)^{g+1-d} \times \mathbb{C}^{N-g+d-1}$?

Here "in bijection" and "identify" refer to the algebraic structures that are used throughout [Ham20]: the bijections of $\mathcal{R}_\Gamma$ with affine spaces in [Ham20, Section 3] are given by polynomial maps with polynomial inverses, and the plane locus in [Ham20, Example 4.6] is identified with $\mathbb{C}^* \times \mathbb{C}^8$ by such a map. The nonempty loci considered below and their proposed products all have the cardinality of the continuum, so a set-theoretic bijection alone would carry no geometric information.

We study the semigroup

$$\Gamma = \langle 10, 11, 34, 46, 58 \rangle, \qquad g = 4,$$

whose conductor is 60. Write $\mathcal{E}_d \subseteq \mathcal{R}_\Gamma$ for the set of subalgebras with embedding dimension exactly d, so that $\mathcal{R}_\Gamma^{(d)} = \bigcup_{k \le d} \mathcal{E}_k$.

Theorem 1.3. *Let $\Gamma = \langle 10, 11, 34, 46, 58 \rangle$. There is an isomorphism of affine varieties*

$$\Phi = (q, A, B, C) \colon \mathcal{R}_\Gamma \xrightarrow{\sim} \mathbb{A}^{54} \times \mathbb{A}^3$$

such that, for every $R \in \mathcal{R}_\Gamma$, the minimum number of generators of R equals

$$\operatorname{edim} R \;=\; 5 - \operatorname{rank} \begin{pmatrix} A(R) & C(R) \\ 0 & B(R) \end{pmatrix}.$$

The same number is the minimum number of topological generators of the corresponding complete subalgebra of $\mathbb{C}[\![t]\!]$. Consequently,

$$\mathcal{R}_\Gamma^{(2)} = \emptyset, \qquad \mathcal{R}_\Gamma^{(3)} = \mathcal{E}_3 \cong \mathbb{A}^{54} \times \{AB \neq 0\},$$
$$\mathcal{R}_\Gamma^{(4)} \cong \mathbb{A}^{54} \times (\mathbb{A}^3 \setminus \{0\}), \qquad \mathcal{R}_\Gamma^{(5)} = \mathcal{R}_\Gamma \cong \mathbb{A}^{57}.$$

The coordinates q consist of 54 of the normal-form coefficients, and A, B, C are coefficients occurring when two binomial relations are reduced; see Theorem 5.3. The exact strata $\mathcal{E}_4$ and $\mathcal{E}_5$ are described in Corollary 5.4.

Corollary 1.4. *Let $\Gamma = \langle 10, 11, 34, 46, 58\rangle$, so that $g = 4$, and take $d = 4$.*

(1) *There is no polynomial $f_4 \in \mathbb{C}[\xi_0, \dots, \xi_4]$ such that $R \in \mathcal{R}_\Gamma^{(4)}$ if and only if $\operatorname{ord}\operatorname{red}_{\langle v_0,\dots,v_3\rangle}(\varphi_R(f_4)) = v_4$. More strongly, $\mathcal{R}_\Gamma^{(4)}$ is an open subvariety of $\mathcal{R}_\Gamma$ which is not of the form $\{h \neq 0\}$ for a regular function h on $\mathcal{R}_\Gamma$, nor a union of two such sets.*

(2) *The space $\mathcal{R}_\Gamma$ is isomorphic to $\mathbb{C}^{57}$, but $\mathcal{R}_\Gamma^{(4)}$ is neither isomorphic as a variety nor homeomorphic in the classical topology to $(\mathbb{C}^*)^{g+1-d} \times \mathbb{C}^{N-g+d-1} = \mathbb{C}^* \times \mathbb{C}^{56}$.*

Thus Question 1.1, in its formulation with fixed polynomials, and Question 1.2, when "identify" means an isomorphism of varieties or even a homeomorphism, have negative answers. The other strata behave as predicted by Question 1.2: $\mathcal{R}_\Gamma^{(3)} \cong (\mathbb{C}^*)^2 \times \mathbb{A}^{55}$ and $\mathcal{R}_\Gamma^{(5)} = \mathcal{R}_\Gamma \cong \mathbb{A}^{57}$; the failure occurs exactly at $d = 4$. The stratum $\mathcal{R}_\Gamma^{(4)}$ is open in $\mathcal{R}_\Gamma$, but it is not principal open, as the remark after Question 1.1 would require; the obstruction is that its complement has codimension three.

The proofs are explicit. Below the conductor, the five normal-form generators produce 31 monomials but only four coincidences of weights, and all of them are governed by the two binomials $H^4 - XZ$ and $H^2Z - XW$ (Lemma 3.2). This yields a complete list of five equations for $\mathcal{R}_\Gamma$ (Proposition 4.2) and a complete list of linear relations in the cotangent space (Proposition 5.1). A weight grading on the coefficient ring, induced by rescaling t, controls which coefficients can occur in which equation (Section 3.2); it allows us to use large polynomials, such as the coordinate C, without expanding them.

We do not attempt a classification of semigroups for which the phenomenon occurs. The methods used to describe $\mathcal{R}_\Gamma$, namely normal forms, truncation at the conductor and reduction of binomial relations, are those of [Ham20]; see also [AGM17] for closely related division and basis criteria for subalgebras of $K[\![x]\!]$.

Organization. Section 2 recalls the setting and relates the number of generators to cotangent spaces, including the truncation at the conductor. Section 3 collects the arithmetic of Γ, the monomials below the conductor and the weight grading. Section 4 determines the equations of $\mathcal{R}_\Gamma$ and proves $\mathcal{R}_\Gamma \cong \mathbb{A}^{57}$. Section 5 computes the cotangent spaces and constructs the coordinates of Theorem 1.3. Section 6 gives an explicit three-parameter family, and Section 7 proves Corollary 1.4. Appendix A lists four of the equations in expanded form.

2. Preliminaries

Throughout we work over $\mathbb{C}$. For $f \in \mathbb{C}[\![t]\!]$ or $f \in \mathbb{C}[t]/(t^c)$ nonzero, $\operatorname{ord} f$ is the smallest exponent with a nonzero coefficient, and f is *monic* if that coefficient is 1. We set $\operatorname{ord} 0 = +\infty$.

2.1. **Complete subalgebras and normal forms.** Let $\Gamma \subseteq \mathbb{N}$ be a numerical semigroup with conductor c, minimally generated by $v_0 < \cdots < v_g$. If $\widehat{R} \subseteq \mathbb{C}[\![t]\!]$ is a complete $\mathbb{C}$-subalgebra whose set of orders is Γ, then $t^c\mathbb{C}[\![t]\!] \subseteq \widehat{R}$ [Ham20, Proposition 2.1]. Hence $\widehat{R}$ is determined by its image R in $J_c = \mathbb{C}[t]/(t^c)$, and $\widehat{R} \mapsto R$ is a bijection onto the set of $\mathbb{C}$-subalgebras $R \subseteq J_c$ with

$$\Gamma_R := \{\operatorname{ord} f \mid 0 \neq f \in R\} \cup \{n \in \mathbb{N} \mid n \geq c\} = \Gamma.$$

Following [Ham20, Section 2] we work with these truncated subalgebras. A tuple $(\eta_0(t), \dots, \eta_g(t))$ in J_c is in *normal form* if each $\eta_i(t)$ is monic of order v_i and all its other exponents with nonzero coefficient are gaps of Γ [Ham20, Definition 2.2]. Every R with $\Gamma_R = \Gamma$ is generated by a unique tuple in normal form [Ham20, Proposition 2.4]. Taking the coefficients of the normal forms as coordinates identifies $\mathcal{R}_\Gamma$ with a Zariski closed subset of $\mathbb{A}^{M(\Gamma)}$, where $M(\Gamma) = \sum_i \#\{\text{gaps} > v_i\}$ [Ham20, Theorem 2.22]. We regard $\mathcal{R}_\Gamma$ as an affine variety with its reduced structure.

Remark 2.1. The group $\operatorname{Aut}\mathbb{C}[\![t]\!]$ of substitutions $t \mapsto \theta(t)$ with $\operatorname{ord}\theta = 1$ acts on $\mathcal{R}_\Gamma$, and its orbits are the analytic equivalence classes [Ham20, Section 4.3]. Since the number of generators is invariant under this action, each stratum $\mathcal{R}_\Gamma^{(d)}$ is a union of orbits.

2.2. **Generators and cotangent spaces.** For $R \in \mathcal{R}_\Gamma$ let $\mathfrak{m} = \{f \in R \mid f = 0 \text{ or } \operatorname{ord} f > 0\}$. Then $R = \mathbb{C} \oplus \mathfrak{m}$, the ideal $\mathfrak{m}$ is nilpotent, and $(R, \mathfrak{m})$ is a local Artinian $\mathbb{C}$-algebra. We put $\operatorname{edim} R = \dim_{\mathbb{C}} \mathfrak{m}/\mathfrak{m}^2$.

Lemma 2.2. *Let $(R, \mathfrak{m})$ be a finite-dimensional local $\mathbb{C}$-algebra with $R = \mathbb{C} \oplus \mathfrak{m}$. The minimum number of elements generating R as a $\mathbb{C}$-algebra is $\dim_{\mathbb{C}} \mathfrak{m}/\mathfrak{m}^2$. Elements $g_1, \dots, g_k \in \mathfrak{m}$ generate R if and only if their classes span $\mathfrak{m}/\mathfrak{m}^2$.*

Proof. Subtracting constants does not change the generated algebra, so we may assume that generators lie in $\mathfrak{m}$. If $R = \mathbb{C}[g_1, \dots, g_k]$ with $g_i \in \mathfrak{m}$, then $\mathfrak{m}$ is spanned by the nonconstant monomials in the g_i, and all of them except the g_i lie in $\mathfrak{m}^2$. Conversely, suppose that the classes of $g_1, \dots, g_k \in \mathfrak{m}$ span $\mathfrak{m}/\mathfrak{m}^2$ and let $S = \mathbb{C}[g_1, \dots, g_k]$ and $L = \operatorname{span}\{g_1, \dots, g_k\}$, so that $\mathfrak{m} = L + \mathfrak{m}^2$. For $j \geq 1$, the space $\mathfrak{m}^j$ is spanned by products $y_1 \cdots y_j$ with $y_i \in \mathfrak{m}$; writing $y_i = \ell_i + n_i$ with $\ell_i \in L$ and $n_i \in \mathfrak{m}^2$, we get $y_1 \cdots y_j \in \ell_1 \cdots \ell_j + \mathfrak{m}^{j+1} \subseteq S + \mathfrak{m}^{j+1}$. Hence $\mathfrak{m}^j \subseteq S + \mathfrak{m}^{j+1}$ for all $j \geq 1$, and therefore $\mathfrak{m} \subseteq S + \mathfrak{m}^j$ for all j. Since $\mathfrak{m}$ is nilpotent, $\mathfrak{m} \subseteq S$. □

Accordingly,

$$\mathcal{R}_\Gamma^{(d)} = \{R \in \mathcal{R}_\Gamma \mid \operatorname{edim} R \leq d\}, \qquad \mathcal{E}_d = \{R \in \mathcal{R}_\Gamma \mid \operatorname{edim} R = d\}.$$

In [Ham20, Definition 4.4], $\mathcal{R}_\Gamma^{(n)}$ is written for $3 \leq n \leq g+1$ and the plane locus $\mathcal{R}_\Gamma^{(2)}$ is denoted $\mathcal{R}_\Gamma^{\mathrm{plane}}$; generation by n elements includes generation by fewer elements.

The next lemma justifies the truncation at the conductor for questions about generators. It applies to our semigroup by Lemma 3.1.

Lemma 2.3. *Assume that every minimal generator of Γ is smaller than c. Let $\widehat{R} \subseteq \mathbb{C}[\![t]\!]$ be a complete subalgebra with set of orders Γ, let $\pi\colon \mathbb{C}[\![t]\!] \to J_c$ be the projection, $R = \pi(\widehat{R})$ and $I = t^c\mathbb{C}[\![t]\!]$. Let $\widehat{\mathfrak{m}} = \{f \in \widehat{R} \mid f = 0 \text{ or } \operatorname{ord} f > 0\}$ and $\mathfrak{m} = \pi(\widehat{\mathfrak{m}})$.*

1. *$I \subseteq \widehat{\mathfrak{m}}^2$, and π induces an isomorphism $\widehat{\mathfrak{m}}/\widehat{\mathfrak{m}}^2 \cong \mathfrak{m}/\mathfrak{m}^2$.*
2. *For $g_1, \dots, g_k \in \widehat{\mathfrak{m}}$, the closure of $\mathbb{C}[g_1, \dots, g_k]$ in the t-adic topology equals $\widehat{R}$ if and only if $\mathbb{C}[\pi(g_1), \dots, \pi(g_k)] = R$.*

In particular, the minimum number of topological generators of $\widehat{R}$ equals $\operatorname{edim} R$.

Proof. Since every minimal generator of Γ is smaller than c, every $n \geq c$ is a sum of two nonzero elements of Γ: otherwise n would be a minimal generator. For each nonzero $a \in \Gamma$ choose a monic $f_a \in \widehat{R}$ of order a.

(1) For $c \le n < c + v_0$ write $n = a + b$ with $a, b \in \Gamma \setminus \{0\}$; then $\mu_n = f_a f_b \in \widehat{\mathfrak{m}}^2$ is monic of order n. Moreover $I \subseteq \widehat{R}$ by [Ham20, Proposition 2.1], hence $I \subseteq \widehat{\mathfrak{m}}$, and $f_{v_0} I = t^{c+v_0}\mathbb{C}[\![t]\!]$ lies in $\widehat{\mathfrak{m}}^2$, because f_{v_0} is t^{v_0} times a unit. Given $f \in I$, we can subtract a linear combination of $\mu_c, \dots, \mu_{c+v_0-1}$ so that the result lies in $t^{c+v_0}\mathbb{C}[\![t]\!]$. Hence $I \subseteq \widehat{\mathfrak{m}}^2$. The map $\widehat{\mathfrak{m}}/\widehat{\mathfrak{m}}^2 \to \mathfrak{m}/\mathfrak{m}^2$ is surjective, and its kernel is $(\pi^{-1}(\mathfrak{m}^2) \cap \widehat{\mathfrak{m}})/\widehat{\mathfrak{m}}^2 = (\widehat{\mathfrak{m}}^2 + I)/\widehat{\mathfrak{m}}^2 = 0$.

(2) Let S be the closure of $\mathbb{C}[g_1, \dots, g_k]$. The set $\widehat{R} = \pi^{-1}(R)$ is a union of cosets of the open subgroup I, hence closed, so $S \subseteq \widehat{R}$. Since I is open, every element of S is congruent modulo I to an element of $\mathbb{C}[g_1, \dots, g_k]$, so $\pi(S) = \mathbb{C}[\pi(g_1), \dots, \pi(g_k)]$. If $S = \widehat{R}$, this gives $R = \mathbb{C}[\pi(g_1), \dots, \pi(g_k)]$. Conversely, assume the latter. Then $\pi(S) = R$, so $S + I = \widehat{R}$, and S contains a monic element s_a of order a for every $a \in \Gamma$ with $a < c$. As in (1), products give monic elements of S of each order in $[c, c + v_0)$, and multiplication by s_{v_0} gives monic elements $s_n \in S$ of every order $n \ge c$. For $f \in I$, successively subtracting multiples of $s_c, s_{c+1}, \dots$ produces partial sums in S which converge to f; since S is a closed subalgebra, $f \in S$. Thus $S = \widehat{R}$.

The final assertion follows from (1), (2) and Lemma 2.2. □

3. The semigroup, monomials and a grading

From now on $\Gamma = \langle 10, 11, 34, 46, 58\rangle$ and $J = \mathbb{C}[t]/(t^{60})$.

Lemma 3.1. *The elements of* Γ *below* 60 *are*

$$0,\ 10,\ 11,\ 20,\ 21,\ 22,\ 30, \dots, 34,\ 40, \dots, 46,\ 50, \dots, 58,$$

twenty-seven in total. Every integer $n \ge 60$ *lies in* Γ *and* $59 \notin \Gamma$*, so the conductor is* $c = 60$*. The gaps are*

$$1, \dots, 9; \quad 12, \dots, 19; \quad 23, \dots, 29; \quad 35, \dots, 39; \quad 47, 48, 49; \quad 59,$$

thirty-three in total. The numbers $10, 11, 34, 46, 58$ *are the minimal generators of* Γ*; all of them are smaller than* c*.*

Proof. The elements $10a + 11b$ below 60 are 0, 10, 11, 20, 21, 22, $30, \dots, 33$, $40, \dots, 44$ and $50, \dots, 55$. Adding 34 to those below 26 gives $34, 44, 45, 54, 55, 56$; adding 46 to those below 14 gives $46, 56, 57$; and 58 is the only remaining element obtained from 58. Sums involving two of $34, 46, 58$ exceed 59. This gives the stated list. The integer 59 is not in the list. The integers $60, \dots, 69$ lie in Γ, as $60 = 6 \cdot 10$, $61 = 50 + 11$, $62 = 40 + 22$, $63 = 30 + 33$, $64 = 20 + 44$, $65 = 10 + 55$, $66 = 6 \cdot 11$, $67 = 33 + 34$, $68 = 34 + 34$ and $69 = 11 + 58$; adding multiples of 10 shows that all $n \ge 60$ lie in Γ. Finally, 34, 46 and 58 are not sums of two smaller nonzero elements of the list, so the five generators are minimal. □

We write $\mathcal{G}$ for the set of gaps and $\Gamma_{<60} = \Gamma \cap [0, 59]$.

3.1. **Normal forms and coordinates.** By Lemma 3.1, a tuple in normal form is

$$\begin{aligned} X &= t^{10} + \sum_{n \in \mathcal{G},\, n > 10} x_n t^n, \qquad H = t^{11} + \sum_{n \in \mathcal{G},\, n > 11} h_n t^n, \\ Z &= t^{34} + \sum_{n \in \mathcal{G},\, n > 34} z_n t^n, \qquad W = t^{46} + \sum_{n \in \mathcal{G},\, n > 46} w_n t^n, \\ U &= t^{58} + u_{59} t^{59}. \end{aligned} \tag{3.1}$$

The numbers of tail coefficients are $24, 24, 9, 4, 1$, so $M(\Gamma) = 62$. Note that there is no coefficient x_{11}, because $11 \in \Gamma$. Let P be the polynomial ring over $\mathbb{C}$ in the 62 coordinates $x_n, h_n, z_n, w_n, u_{59}$, so that $\operatorname{Spec} P = \mathbb{A}^{62}$. We read (3.1) in $J_P = P[t]/(t^{60})$, and for a point $p \in \mathbb{A}^{62}$ we write $X_p, \dots, U_p \in J$ for the specialized tuple and

$$R_p = \mathbb{C}[X_p, H_p, Z_p, W_p, U_p] \subseteq J.$$

By the results recalled in Section 2, $\mathcal{R}_\Gamma$ is identified with the set of points p such that $\Gamma_{R_p} = \Gamma$.

For $\alpha \in \mathbb{N}^5$ put $M^\alpha = X^{\alpha_1} H^{\alpha_2} Z^{\alpha_3} W^{\alpha_4} U^{\alpha_5}$ and $\omega(\alpha) = 10\alpha_1 + 11\alpha_2 + 34\alpha_3 + 46\alpha_4 + 58\alpha_5$, the weighted order. Since each generator has leading term t^v, the monomial M^α is monic of order $\omega(\alpha)$ at every point when $\omega(\alpha) < 60$, and it vanishes in J when $\omega(\alpha) \geq 60$.

Lemma 3.2. *There are exactly* 31 *monomials M^α with $\omega(\alpha) < 60$, including* 1. *Their weighted orders are precisely the* 27 *elements of $\Gamma_{<60}$. Each element of $\Gamma_{<60}$ is the weighted order of exactly one of them, except*

$$44: \ H^4,\ XZ; \qquad 54: \ XH^4,\ X^2Z; \qquad 55: \ H^5,\ XHZ; \qquad 56: \ H^2Z,\ XW.$$

Proof. The monomials in X and H of degree k have weighted orders $10k, \dots, 11k$, one each; for $k \leq 5$ this gives 21 monomials with orders 0; $10, 11$; $20, 21, 22$; $30, \dots, 33$; $40, \dots, 44$; $50, \dots, 55$, and for $k \geq 6$ the order is at least 60. Monomials containing Z once are Z times $1, X, H, X^2, XH, H^2$, of orders $34, 44, 45, 54, 55, 56$. Monomials containing W once are W, XW, HW, of orders $46, 56, 57$. Finally there is U. Every other monomial has order at least 60. This gives $21 + 6 + 3 + 1 = 31$ monomials, and comparing orders gives the stated coincidences. □

Definition 3.3. For $k \in \Gamma_{<60}$ let b_k be the unique monomial of weighted order k in Lemma 3.2, except that we choose

$$b_{44} = XZ, \qquad b_{54} = X^2Z, \qquad b_{55} = XHZ, \qquad b_{56} = XW.$$

Put

$$F = H^4 - XZ, \qquad G = H^2Z - XW.$$

Then $XH^4 - X^2Z = XF$ and $H^5 - XHZ = HF$. In particular $b_0 = 1$, $b_{10} = X$, $b_{11} = H$, $b_{34} = Z$, $b_{45} = HZ$, $b_{46} = W$, $b_{57} = HW$ and $b_{58} = U$. Every b_k with $k \in \Gamma_{<60} \setminus \{0, 10, 11, 34, 46, 58\}$ is a monomial of degree at least two.

Lemma 3.4. *The P-module J_P is free with basis $\{b_k \mid k \in \Gamma_{<60}\} \cup \{t^n \mid n \in \mathcal{G}\}$. Thus every $f \in J_P$ has a unique decomposition*

$$f = \sum_{k \in \Gamma_{<60}} c_k(f)\, b_k + \rho(f), \qquad c_k(f) \in P, \quad \rho(f) = \sum_{n \in \mathcal{G}} \rho_n(f)\, t^n, \ \rho_n(f) \in P,$$

which is computed by eliminating, in increasing order of $k \in \Gamma_{<60}$, the coefficient of t^k with the monic element b_k. If all coefficients of f below t^m vanish, then $c_k(f) = 0$ for $k < m$. The decomposition commutes with specialization at any point $p \in \mathbb{A}^{62}$, where it is the unique decomposition of $f(p)$ with respect to the basis $\{b_k(p)\} \cup \{t^n\}$ of J.

Proof. The 60 elements are monic of the 60 distinct orders $0, \dots, 59$, so the transition matrix to $1, t, \dots, t^{59}$ is unitriangular over P. The elimination uses only ring operations in P, since each b_k is monic, and it commutes with evaluation at p. □

We call $\rho(f)$ the *remainder* of f. The polynomials $c_k(f)$ and $\rho_n(f)$ are regular functions on $\mathbb{A}^{62}$.

3.2. **A weight grading.** Grade P by assigning to each coordinate the difference between its exponent and the order of its generator:

$$\operatorname{wt} x_n = n-10, \quad \operatorname{wt} h_n = n-11, \quad \operatorname{wt} z_n = n-34, \quad \operatorname{wt} w_n = n-46, \quad \operatorname{wt} u_{59} = 1.$$

All weights are positive; the smallest ones are $\operatorname{wt} h_{12} = \operatorname{wt} z_{35} = \operatorname{wt} w_{47} = \operatorname{wt} u_{59} = 1$ and $\operatorname{wt} x_{12} = 2$. This grading corresponds to the rescaling $t \mapsto \lambda t$ followed by renormalization of the leading coefficients, which multiplies each coordinate y by $\lambda^{\operatorname{wt} y}$.

Lemma 3.5. *A polynomial is homogeneous of degree δ if it is a linear combination of monomials of degree δ; the zero polynomial is homogeneous of every degree, and a nonzero polynomial is never homogeneous of negative degree.*

1. *For every α, the coefficient of t^n in M^α is homogeneous of degree $n-\omega(\alpha)$.*
2. *Let $f \in J_P$ and $\nu \in \mathbb{Z}$ be such that the coefficient of t^n in f is homogeneous of degree $n-\nu$ for every n. Then $c_k(f)$ is homogeneous of degree $k-\nu$, and $\rho_n(f)$ is homogeneous of degree $n-\nu$.*

Proof. (1) The coefficient of t^n in a generator of order v is 1 for $n = v$, and otherwise it is 0 or a coordinate of weight $n-v$. For a product, the coefficient of t^n is $\sum_{i+j=n} a_i b_j$, which is homogeneous of degree $(i-\omega)+(j-\omega') = n-\omega-\omega'$. (2) The property is preserved by the elimination of Lemma 3.4: at step k, the coefficient $c_k(f)$ is the current coefficient of t^k, of degree $k-\nu$, and subtracting $c_k(f)b_k$ changes the coefficient of t^n by a polynomial of degree $(k-\nu)+(n-k) = n-\nu$ by (1). □

Lemma 3.6. *Let $Q \in P$ be homogeneous of degree $\delta \geq 1$ and let y be a coordinate.*

1. *If $\operatorname{wt} y > \delta$, then Q does not involve y.*
2. *If $\operatorname{wt} y = \delta$, then $Q = \gamma y + Q'$ with $\gamma \in \mathbb{C}$ and Q' not involving y. Moreover γ is the value of Q at the point where $y = 1$ and all other coordinates vanish.*

Proof. Since all weights are positive, a monomial containing y has degree at least $\operatorname{wt} y$, with equality only for y itself. The monomials of Q' have positive degree in the other coordinates and therefore vanish at the indicated point. □

We shall use Lemma 3.6(2) at points where only one coordinate is nonzero. At such a point four of the five generators are monomials t^v, and the decompositions of Lemma 3.4 can be read off directly.

4. The equations of the parameter space

Lemma 4.1. *For every $p \in \mathbb{A}^{62}$, the algebra R_p is the linear span of the 31 monomials of Lemma 3.2 evaluated at p, and*

$$R_p = V_p + \operatorname{span}\{F_p,\ X_pF_p,\ H_pF_p,\ G_p\}, \qquad V_p = \operatorname{span}\{b_k(p) \mid k \in \Gamma_{<60}\}.$$

Moreover $\dim V_p = 27$, and the orders of the nonzero elements of V_p are exactly the elements of $\Gamma_{<60}$.

Proof. R_p is spanned by all monomials in the generators, and those of weighted order at least 60 vanish in J. The four monomials not chosen in Definition 3.3 are $H^4 = b_{44} + F$, $XH^4 = b_{54} + XF$, $H^5 = b_{55} + HF$ and $H^2Z = b_{56} + G$. The elements $b_k(p)$ are monic of distinct orders k, so they are linearly independent, and a nonzero linear combination has the order of its first nonzero term. □

Proposition 4.2. *A point $p \in \mathbb{A}^{62}$ lies in $\mathcal{R}_\Gamma$ if and only if $\rho(F)(p) = 0$ and $\rho(G)(p) = 0$. In this case $R_p = V_p$.*

Proof. By Lemma 4.1, $V_p \subseteq R_p$ already realizes every order in $\Gamma_{<60}$, so $p \in \mathcal{R}_\Gamma$ if and only if no nonzero element of R_p has order in $\mathcal{G}$.

Assume $p \in \mathcal{R}_\Gamma$. The element $\rho(F)(p) = F_p - \sum_k c_k(F)(p)\, b_k(p)$ lies in R_p and its nonzero coefficients sit at gaps; if it were nonzero, its order would be a gap. Hence $\rho(F)(p) = 0$, and likewise $\rho(G)(p) = 0$.

Conversely, assume that both remainders vanish at p; we suppress p from the notation. Then $F, G \in V$. The coefficients of t^n in F vanish for $n \le 44$, because H^4 and XZ are both monic of order 44; hence $c_k(F) = 0$ for $k \le 44$ by Lemma 3.4. Since $\Gamma \cap [45, 59] = \{45, 46, 50, \dots, 58\}$,

$$F = c_{45}(F)\, HZ + c_{46}(F)\, W + \sum_{k=50}^{58} c_k(F)\, b_k.$$

Multiplying by X and by H, the products Xb_k and Hb_k with $k \ge 50$ have weighted order at least 60 and vanish. Therefore

$$XF = c_{45}(F)\, b_{55} + c_{46}(F)\, b_{56}, \qquad HF = c_{45}(F)\, (b_{56} + G) + c_{46}(F)\, b_{57},$$

using $XHZ = b_{55}$, $XW = b_{56}$, $H^2Z = b_{56} + G$ and $HW = b_{57}$. Both lie in V. By Lemma 4.1, $R = V$, and the orders of its nonzero elements are exactly $\Gamma_{<60}$. □

Remark 4.3. Proposition 4.2 is an instance of the criterion of [Ham20, Proposition 2.17 and Theorem 2.22], with a different choice of the monomials used for reduction, and of the binomial criterion for bases in [AGM17, Theorem 2.9]. We have included the short proof because the identification $R_p = V_p$ and the explicit form of the relations are used below. Since both descriptions characterize the same subset of $\mathbb{A}^{62}$, they define the same reduced variety $\mathcal{R}_\Gamma$.

4.1. **The low-order terms.** We now make the remainders explicit. Write f_n for the coefficient of t^n in $f \in J_P$, and introduce the names

$$\delta = c_{45}(F), \quad A = c_{46}(F), \quad C = c_{58}(F), \quad \varepsilon = c_{57}(G), \quad B = c_{58}(G).$$

Proposition 4.4. *The remainders are*

$$\rho(F) = r_{47}t^{47} + r_{48}t^{48} + r_{49}t^{49} + r_{59}t^{59}, \qquad \rho(G) = q_{59}t^{59},$$

where

$$\delta = 4h_{12} - z_{35}, \tag{4.1}$$

$$A = 4h_{13} + 6h_{12}^2 - x_{12} - z_{36} - \delta\,(h_{12} + z_{35})$$

$$= 4h_{13} + 2h_{12}^2 - 3h_{12}z_{35} + z_{35}^2 - x_{12} - z_{36}, \tag{4.2}$$

$$\varepsilon = 2h_{12} + z_{35} - w_{47}, \tag{4.3}$$

$$B = 2h_{13} + h_{12}^2 + 2h_{12}z_{35} + z_{36} - x_{12} - w_{48} - \varepsilon\,(h_{12} + w_{47})$$

$$= 2h_{13} - h_{12}^2 + h_{12}z_{35} - h_{12}w_{47} - z_{35}w_{47} + w_{47}^2 + z_{36} - x_{12} - w_{48}, \tag{4.4}$$

and

$$r_n = F_n - \delta\,(HZ)_n - A\,w_n \qquad (n = 47, 48, 49), \tag{4.5}$$

$$q_{59} = G_{59} - \varepsilon\,(HW)_{59} - B\,u_{59}. \tag{4.6}$$

The polynomials $c_k(F)$ and $\rho_n(F)$ are homogeneous of degrees $k-44$ and $n-44$, and $c_k(G)$, $\rho_n(G)$ are homogeneous of degrees $k-56$ and $n-56$. In particular

$$\deg A = \deg B = 2, \qquad \deg C = 14, \qquad \deg q_{59} = 3,$$
$$\deg r_{47} = 3, \qquad \deg r_{48} = 4, \qquad \deg r_{49} = 5, \qquad \deg r_{59} = 15.$$

Proof. By Lemma 3.5(1) the coefficients of F and G are homogeneous of degree $n-44$ and $n-56$, so the degree statements follow from Lemma 3.5(2). We have $F_n = 0$ for $n \le 44$ and $G_n = 0$ for $n \le 56$, as both terms of each binomial are monic of the same order. The gaps above 44 are $47, 48, 49, 59$ and the only gap above 56 is 59, which gives the shape of the remainders. Writing

$$X = t^{10}(1 + x_{12}t^2 + \cdots), \qquad H = t^{11}(1 + h_{12}t + h_{13}t^2 + \cdots),$$
$$Z = t^{34}(1 + z_{35}t + z_{36}t^2 + \cdots),$$

we get

$$F = t^{44}\big((4h_{12} - z_{35})t + (4h_{13} + 6h_{12}^2 - x_{12} - z_{36})t^2 + \cdots\big),$$
$$HZ = t^{45}\big(1 + (h_{12} + z_{35})t + \cdots\big).$$

The elimination of Lemma 3.4 first subtracts $\delta\, HZ$ with $\delta = F_{45}$, then $A\,W$ with A the resulting coefficient of t^{46}; this is (4.1) and (4.2). The next element of Γ is 50, so the coefficients of t^{47}, t^{48}, t^{49} in $F - \delta HZ - AW$ are never changed again; since $W_n = w_n$ for $n = 47, 48, 49$, this is (4.5). Similarly

$$H^2Z = t^{56}\big(1 + (2h_{12} + z_{35})t + (2h_{13} + h_{12}^2 + 2h_{12}z_{35} + z_{36})t^2 + \cdots\big),$$
$$XW = t^{56}\big(1 + w_{47}t + (x_{12} + w_{48})t^2 + \cdots\big),$$
$$HW = t^{57}\big(1 + (h_{12} + w_{47})t + \cdots\big),$$

and the elimination subtracts $\varepsilon\, HW$ and $B\,U$, giving (4.3), (4.4) and (4.6), because $U_{59} = u_{59}$. □

The expansions of r_{47}, r_{48}, r_{49} and q_{59}, with 17, 22, 26 and 24 terms, are listed in Appendix A. The polynomials r_{59} and C are large, and we shall use only the structural information of Proposition 4.4 and Lemma 3.6 about them.

4.2. **Solving the equations.** Call the 57 coordinates other than

$$y_1 = z_{37}, \quad y_2 = z_{38}, \quad y_3 = w_{49}, \quad y_4 = z_{39}, \quad y_5 = z_{49}$$

the *free coordinates*. They are the 48 coefficients of X and H together with

$$z_{35},\ z_{36},\ z_{47},\ z_{48},\ z_{59},\ w_{47},\ w_{48},\ w_{59},\ u_{59}.$$

Order the five equations as

$$e_1 = r_{47}, \quad e_2 = r_{48}, \quad e_3 = q_{59}, \quad e_4 = r_{49}, \quad e_5 = r_{59}.$$

Theorem 4.5. *For $k = 1, \dots, 5$ we have $\deg e_k = \operatorname{wt} y_k$, and*

$$e_k = -y_k + Q_k, \qquad Q_k \in \mathbb{C}[\textit{free coordinates}, y_1, \dots, y_{k-1}].$$

Consequently there are homogeneous polynomials P_k in the free coordinates, of degree $\operatorname{wt} y_k$, such that

$$\mathcal{R}_\Gamma = \{p \in \mathbb{A}^{62} \mid y_k(p) = P_k(p) \text{ for } k = 1, \dots, 5\},$$

and the projection to the free coordinates is an isomorphism $\mathcal{R}_\Gamma \cong \mathbb{A}^{57}$.

Proof. The weights are $\operatorname{wt} z_{37} = 3$, $\operatorname{wt} z_{38} = 4$, $\operatorname{wt} w_{49} = 3$, $\operatorname{wt} z_{39} = 5$, $\operatorname{wt} z_{49} = 15$, which equal the degrees $3, 4, 3, 5, 15$ of Proposition 4.4. By Lemma 3.6(2), $e_k = \gamma_k y_k + Q'_k$ with Q'_k not involving y_k, and γ_k is obtained by evaluating at the point where $y_k = 1$ and all other coordinates vanish.

At the point $z_n = 1$ with $n \in \{37, 38, 39, 49\}$ we have $X = t^{10}$, $H = t^{11}$, $Z = t^{34} + t^n$, so $F = -t^{n+10}$, which is a gap monomial for $n + 10 \in \{47, 48, 49, 59\}$. Hence $\rho(F) = -t^{n+10}$ and the coefficient of y_k in r_{n+10} is -1 for $k = 1, 2, 4, 5$. At the point $w_{49} = 1$ we have $W = t^{46} + t^{49}$ and all other generators monomial, so $G = t^{56} - (t^{56} + t^{59}) = -t^{59}$, and the coefficient of y_3 in q_{59} is -1.

It remains to show that e_k does not involve $y_{k+1}, \dots, y_5$. Since $\operatorname{wt} z_{38} = 4 > \deg e_1$, $\operatorname{wt} z_{39} = 5 > \deg e_1, \deg e_2, \deg e_3$, and $\operatorname{wt} z_{49} = 15 > \deg e_1, \dots, \deg e_4$, Lemma 3.6(1) leaves only the occurrence of w_{49} in $e_1 = r_{47}$ and $e_2 = r_{48}$. In (4.5), the polynomials F_n, $(HZ)_n$, δ and A involve no coefficient of W, so the only coefficient of W in r_n is w_n. Hence w_{49} occurs in neither r_{47} nor r_{48}.

By Proposition 4.2, $\mathcal{R}_\Gamma$ is the zero set of $e_1, \dots, e_5$. The system $y_k = Q_k$ is triangular: substituting recursively gives polynomials P_k in the free coordinates, homogeneous of degree $\operatorname{wt} y_k$ because the Q_k are, and $\mathcal{R}_\Gamma$ is the graph of the polynomial map $(P_1, \dots, P_5)$. The projection to the free coordinates is therefore an isomorphism, with inverse given by the P_k. □

Remark 4.6. Put $I_k = (e_1, \dots, e_k)$ and $I'_k = (y_1 - P_1, \dots, y_k - P_k)$ in P. If $I_{k-1} = I'_{k-1}$, then $Q_k \equiv P_k$ modulo I'_{k-1}, because P_k is obtained from Q_k by substituting P_j for y_j, $j < k$; hence $e_k \equiv -(y_k - P_k)$ and $I_k = I'_k$. By induction $I_5 = I'_5$, and P/I'_5 is the polynomial ring in the 57 free coordinates. Thus the closed subscheme of $\mathbb{A}^{62}$ defined by $e_1, \dots, e_5$ is reduced and isomorphic to $\mathbb{A}^{57}$.

5. Cotangent spaces and global coordinates

Proposition 5.1. *Let $p \in \mathcal{R}_\Gamma$ and $R = R_p$. In $\mathfrak{m}/\mathfrak{m}^2$ the classes of X, H, Z, W, U span, and the space of linear relations among them is spanned by*

$$A\,[W] + C\,[U] = 0 \qquad \text{and} \qquad B\,[U] = 0,$$

with A, B, C evaluated at p. Consequently

$$\operatorname{edim} R = 5 - \operatorname{rank} \begin{pmatrix} A & C \\ 0 & B \end{pmatrix}.$$

Proof. Let $\Gamma_2 = \Gamma_{<60} \setminus \{0, 10, 11, 34, 46, 58\}$, a set of 21 elements, and $N = \operatorname{span}\{b_k \mid k \in \Gamma_2\}$. By Proposition 4.2, $R = V$ has basis $\{b_k\}$, so

$$\mathfrak{m} = N \oplus \operatorname{span}\{X, H, Z, W, U\}.$$

The ideal $\mathfrak{m}^2$ is spanned by the products of two nonconstant monomials, that is, by the 25 monomials of degree at least two among the 31 monomials of Lemma 3.2. These are the 21 elements b_k, $k \in \Gamma_2$, together with $H^4 = b_{44} + F$, $XH^4 = b_{54} + XF$, $H^5 = b_{55} + HF$ and $H^2 Z = b_{56} + G$. Hence

$$\mathfrak{m}^2 = N + \operatorname{span}\{F,\ XF,\ HF,\ G\}.$$

We compute these four elements modulo N, using the decompositions in the proof of Proposition 4.2. Since $45, 50, \dots, 57 \in \Gamma_2$,

$$\begin{aligned} F &\equiv A\,W + C\,U, & G &= \varepsilon\, b_{57} + B\,U \equiv B\,U, \\ XF &\equiv 0, & HF &\equiv \delta\, G \equiv \delta B\,U \qquad (\operatorname{mod} N). \end{aligned}$$

Therefore $\mathfrak{m}^2 = N \oplus \operatorname{span}\{AW + CU,\ BU\}$, and

$$\mathfrak{m}/\mathfrak{m}^2 \cong \operatorname{span}\{X, H, Z, W, U\} \big/ \operatorname{span}\{AW + CU,\ BU\}.$$

The dimension of the denominator is the rank of $\left(\begin{smallmatrix} A & C \\ 0 & B \end{smallmatrix}\right)$. □

In particular $\operatorname{edim} R \geq 3$ for every $R \in \mathcal{R}_\Gamma$, so $\mathcal{R}_\Gamma^{(2)} = \emptyset$: by Lemma 2.3, none of the corresponding complete subalgebras of $\mathbb{C}[\![t]\!]$ is the local ring of a plane branch. This is consistent with the criterion recalled in [Ham20, Section 4.2], which fails for Γ already because $\gcd(10, 11) = 1$.

Lemma 5.2. *On $\mathcal{R}_\Gamma$, expressed in the free coordinates,*

$$A = a - z_{36}, \qquad B = b - w_{48}, \qquad C = -z_{48} + \Psi,$$

where a is a polynomial in x_{12}, h_{12}, h_{13}, z_{35}; b is a polynomial in x_{12}, h_{12}, h_{13}, z_{35}, z_{36}, w_{47}; and Ψ is a polynomial in the free coordinates other than z_{48}.

Proof. The statements about A and B are (4.2) and (4.4); these formulas involve no eliminated coordinate. The polynomial $C = c_{58}(F) \in P$ is homogeneous of degree $14 = \operatorname{wt} z_{48}$. By Lemma 3.6, it does not involve z_{49}, of weight 15, and $C = \gamma z_{48} + C'$ with C' free of z_{48}. At the point where $z_{48} = 1$ and all other coordinates vanish, $F = -t^{58} = -U$, so $\gamma = -1$. On $\mathcal{R}_\Gamma$ we substitute $y_k = P_k$ for $k \leq 4$; these P_k have degrees $3, 4, 3, 5 < 14$, so they do not involve z_{48} by Lemma 3.6(1). Hence $\Psi := C'|_{y_k = P_k}$ does not involve z_{48}. □

Theorem 5.3. *Let q denote the 54 free coordinates other than z_{36}, w_{48}, z_{48}, namely the 48 coefficients of X and H together with $z_{35}, z_{47}, z_{59}, w_{47}, w_{59}, u_{59}$. Then*

$$\Phi = (q, A, B, C) \colon \mathcal{R}_\Gamma \longrightarrow \mathbb{A}^{54} \times \mathbb{A}^3$$

is an isomorphism of affine varieties, and $\operatorname{edim} R = 5 - \operatorname{rank}\left(\begin{smallmatrix} A & C \\ 0 & B \end{smallmatrix}\right)$ for every $R \in \mathcal{R}_\Gamma$.

Proof. By Theorem 4.5 it suffices to show that $(q, z_{36}, w_{48}, z_{48}) \mapsto (q, A, B, C)$ is an automorphism of $\mathbb{A}^{57}$. By Lemma 5.2 it is given by

$$A = a(q) - z_{36}, \qquad B = b(q, z_{36}) - w_{48}, \qquad C = -z_{48} + \Psi(q, z_{36}, w_{48}),$$

and its inverse is

$$z_{36} = a(q) - A, \qquad w_{48} = b\big(q, a(q) - A\big) - B, \qquad z_{48} = \Psi(q, z_{36}, w_{48}) - C,$$

with z_{36} and w_{48} substituted in the last formula. Both maps are polynomial. The formula for edim is Proposition 5.1. □

Corollary 5.4. *Under Φ, the embedding dimension strata are as follows.*

d	*exactly d:* $\mathcal{E}_d$	*at most d:* $\mathcal{R}_\Gamma^{(d)}$
≤ 2	$\emptyset$	$\emptyset$
3	$\mathbb{A}^{54} \times \{AB \neq 0\}$	$\mathbb{A}^{54} \times \{AB \neq 0\}$
4	$\mathbb{A}^{54} \times \big(\{AB = 0\} \setminus \{0\}\big)$	$\mathbb{A}^{54} \times (\mathbb{A}^3 \setminus \{0\})$
5	$\mathbb{A}^{54} \times \{0\}$	$\mathbb{A}^{54} \times \mathbb{A}^3$

In particular $\mathcal{R}_\Gamma^{(3)} \cong (\mathbb{C}^*)^2 \times \mathbb{A}^{55}$ is dense in $\mathcal{R}_\Gamma$, $\mathcal{E}_5 \cong \mathbb{A}^{54}$ is closed of codimension three, and $\mathcal{R}_\Gamma^{(4)}$ is a smooth irreducible open subset of $\mathcal{R}_\Gamma$. The stratum $\mathcal{E}_4$ has two irreducible components, $\mathbb{A}^{54} \times (\{A = 0\} \setminus \{0\})$ and $\mathbb{A}^{54} \times (\{B = 0\} \setminus \{0\})$, each of

dimension 56*; they meet along* $\mathbb{A}^{54} \times \{A = B = 0,\ C \neq 0\}$*, of dimension* 55*, which is the singular locus of* $\mathcal{E}_4$*.*

Proof. The matrix $\left(\begin{smallmatrix} A & C \\ 0 & B \end{smallmatrix}\right)$ has determinant AB; it has rank 2 if $AB \neq 0$, rank 0 if $A = B = C = 0$, and rank 1 otherwise. The remaining assertions concern the subsets $\{AB \neq 0\}$, $\{AB = 0\} \setminus \{0\}$ and $\{0\}$ of $\mathbb{A}^3$, the second of which is the union of two coordinate planes without the origin. □

Theorem 1.3 follows from Theorems 4.5 and 5.3, Corollary 5.4 and Lemmas 2.2 and 2.3.

6. An explicit three-parameter family

The following family is a closed subvariety $\mathbb{A}^3 \subseteq \mathcal{R}_\Gamma$ which meets each of the nonempty strata $\mathcal{E}_3$, $\mathcal{E}_4$, $\mathcal{E}_5$. It also gives an independent check of the formulas for A, B, C, through identities which hold exactly in $\mathbb{C}[\![t]\!]$.

Example 6.1. For $(s_1, s_2, s_3) \in \mathbb{C}^3$ put

$$X = t^{10}, \qquad H = t^{11}, \qquad Z = t^{34} + s_1 t^{36} + s_1 s_2\, t^{38} + s_3\, t^{48},$$
$$W = t^{46} + s_2\, t^{48}, \qquad U = t^{58}.$$

This tuple is in normal form, since $36, 38, 48$ are gaps larger than 34 and 48 is a gap larger than 46. In $\mathbb{C}[\![t]\!]$ one has

$$F = H^4 - XZ = -s_1 W - s_3 U, \qquad G = H^2 Z - XW = (s_1 - s_2)\, U + s_1 s_2\, X^6 + s_3 X^7.$$

Modulo t^{60} the last two terms vanish, so $\rho(F) = \rho(G) = 0$ and the point, which we denote by $\kappa(s_1, s_2, s_3)$, lies in $\mathcal{R}_\Gamma$ by Proposition 4.2. By uniqueness of the decomposition, $c_{46}(F) = -s_1$, $c_{58}(F) = -s_3$, $c_{57}(G) = 0$ and $c_{58}(G) = s_1 - s_2$, that is,

$$(A, B, C)\big(\kappa(s_1, s_2, s_3)\big) = (-s_1,\ s_1 - s_2,\ -s_3),$$

in agreement with (4.2) and (4.4). All coordinates of $\kappa(s_1, s_2, s_3)$ in q vanish, so $\Phi \circ \kappa(s_1, s_2, s_3) = (0, -s_1, s_1 - s_2, -s_3)$ and κ is an isomorphism of $\mathbb{A}^3$ onto the closed subvariety $\Phi^{-1}(\{0\} \times \mathbb{A}^3)$. By Theorem 5.3,

$$\operatorname{edim} \kappa(s_1, s_2, s_3) = \begin{cases} 3, & s_1(s_1 - s_2) \neq 0, \\ 5, & s_1 = s_2 = s_3 = 0, \\ 4, & \text{otherwise.} \end{cases}$$

For example, $\kappa(1, 0, 0)$ is generated by X, H, Z, since $W = XZ - H^4$ and $U = H^2 Z - XW$; the point $\kappa(1, 1, 0)$ has embedding dimension four, with $W = XZ - H^4$; and $\kappa(0, 0, 0)$ is the monomial algebra.

At $\kappa(1, 1, 0)$ the orders $10, 11, 34, 46, 58$ of the normal-form generators minimally generate Γ, but the generators themselves do not minimally generate the algebra. This is an instance of the phenomenon noted in [Ham20, Remark 2.5].

7. Hamilton's questions

7.1. **Question 1.1.** Let $T = \langle 10, 11, 34, 46 \rangle$. By Lemma 3.1, $T \cap [0, 57] = \Gamma \cap [0, 57]$ and $58 \notin T$. For $f \in \mathbb{C}[\xi_0, \ldots, \xi_4]$ and $p \in \mathcal{R}_\Gamma$, let $\varphi_p(f) = f(X_p, H_p, Z_p, W_p, U_p) \in J$. The reduction $\operatorname{red}_{\langle v_0, \ldots, v_3 \rangle}$ of [Ham20, Algorithm 2.15] proceeds as follows: for each $n \in T$ with $0 < n < 60$ a monic monomial m_n in $\xi_0, \ldots, \xi_3$ of weighted degree n is fixed, and, in increasing order of n, the coefficient of t^n is removed by

subtracting the appropriate multiple of $m_n(X_p, H_p, Z_p, W_p)$. The argument below uses only these features and not the particular choice of m_n.

Lemma 7.1. *Let $f \in \mathbb{C}[\xi_0, \dots, \xi_4]$ with $f(0) = 0$, and put $\sigma_p = \operatorname{red}_{\langle v_0,\dots,v_3\rangle}(\varphi_p(f))$ for $p \in \mathcal{R}_\Gamma$. There is a polynomial $h_f \in P$ such that, for every $p \in \mathcal{R}_\Gamma$,*

$$\operatorname{ord}\sigma_p = 58 \iff h_f(p) \neq 0.$$

If $f(0) \neq 0$, then $\operatorname{ord}\sigma_p = 0$ for every $p \in \mathcal{R}_\Gamma$.

Proof. Each $m_n(X, H, Z, W)$ is monic of order n with coefficients in P, so every step of the reduction is a polynomial operation, and the coefficient h_f of t^{58} in the output is a polynomial in the coordinates. The constant term of $\varphi_p(f)$ is $f(0)$, and the reduction only subtracts elements of positive order; this proves the last assertion. Assume $f(0) = 0$. Then $\sigma_p \in R_p$ has no constant term, and its coefficients at all $n \in T \cap [1, 59]$ vanish, since later steps only change higher coefficients. As $p \in \mathcal{R}_\Gamma$, a nonzero σ_p has order in $\Gamma \cap [1, 59]$. The elements of $\Gamma \cap [1, 57]$ lie in T, and 59 is a gap; hence either $\sigma_p = 0$ or $\operatorname{ord}\sigma_p = 58$, and the latter holds exactly when $h_f(p) \neq 0$. □

Lemma 7.2. *No polynomial $\phi \in \mathbb{C}[s_1, s_2]$ vanishes at exactly one point of $\mathbb{C}^2$.*

Proof. Suppose $\phi(a, b) = 0$, so ϕ is not a nonzero constant. If $\phi = 0$ there is nothing to prove. If ϕ involves s_2, write $\phi = \sum_{j=0}^{m} \phi_j(s_1) s_2^j$ with $m \geq 1$ and $\phi_m \neq 0$; for each of the infinitely many s_1 with $\phi_m(s_1) \neq 0$ there is a root s_2. If ϕ does not involve s_2, then $\phi(a, s_2) = 0$ for all s_2. □

Theorem 7.3. *Let $\Gamma = \langle 10, 11, 34, 46, 58\rangle$.*

(1) *There is no polynomial $f \in \mathbb{C}[\xi_0, \dots, \xi_4]$ such that, for all $p \in \mathcal{R}_\Gamma$,*

$$p \in \mathcal{R}_\Gamma^{(4)} \iff \operatorname{ord}\operatorname{red}_{\langle v_0,\dots,v_3\rangle}(\varphi_p(f)) = 58.$$

(2) *There is no regular function h on $\mathcal{R}_\Gamma$ with $\mathcal{R}_\Gamma^{(4)} = \{h \neq 0\}$.*
(3) *There are no regular functions h_1, h_2 on $\mathcal{R}_\Gamma$ such that*

$$\mathcal{R}_\Gamma^{(4)} = \{h_1 \neq 0\} \cup \{h_2 \neq 0\}.$$

On the other hand, $\mathcal{R}_\Gamma^{(4)} = \{A \neq 0\} \cup \{B \neq 0\} \cup \{C \neq 0\}$.

Proof. (2) Let h be a regular function on $\mathcal{R}_\Gamma$ and consider the plane $\lambda(s_1, s_2) = \kappa(s_1, s_2, 0)$ of Example 6.1. Then $h \circ \lambda \in \mathbb{C}[s_1, s_2]$, and by Example 6.1, $\lambda(s_1, s_2) \in \mathcal{R}_\Gamma^{(4)}$ if and only if $(s_1, s_2) \neq (0, 0)$. If $\mathcal{R}_\Gamma^{(4)} = \{h \neq 0\}$, then $h \circ \lambda$ vanishes exactly at the origin, contradicting Lemma 7.2.

(1) If $f(0) \neq 0$, the condition never holds by Lemma 7.1, while $\mathcal{R}_\Gamma^{(4)} \neq \emptyset$. If $f(0) = 0$, Lemma 7.1 gives $\{p \mid \operatorname{ord}\sigma_p = 58\} = \{h_f \neq 0\}$, which differs from $\mathcal{R}_\Gamma^{(4)}$ by (2).

(3) Under Φ, the complement of $\mathcal{R}_\Gamma^{(4)}$ is $\mathcal{E}_5 = \mathbb{A}^{54} \times \{0\}$, which is irreducible and nonempty of codimension three in $\mathbb{A}^{57}$. The complement of $\{h_1 \neq 0\} \cup \{h_2 \neq 0\}$ is the zero set of h_1, h_2. If it is nonempty, each of its irreducible components corresponds to a minimal prime over (h_1, h_2) in the polynomial ring $\mathbb{C}[\mathbb{A}^{57}]$, which has height at most two by Krull's height theorem [Mat86, Theorem 13.5]; hence each component has codimension at most two. This is impossible.

Finally, the description of $\mathcal{R}_\Gamma^{(4)}$ by A, B, C is Corollary 5.4. □

Theorem 7.3 proves Corollary 1.4(1). It concerns the formulation of Question 1.1 with polynomials $f_d, \dots, f_g$ fixed independently of R. By Theorem 7.3(2) and (3), at least three principal open subsets are needed to cover $\mathcal{R}_\Gamma^{(4)}$, and the coordinates A, B, C show that three suffice.

7.2. **Question 1.2.**

Lemma 7.4. *Let $m \geq 0$, let $\mathbb{C}[q, A, B, C]$ be the coordinate ring of $\mathbb{A}^m \times \mathbb{A}^3$, and let $\mathcal{V} = \mathbb{A}^m \times (\mathbb{A}^3 \setminus \{0\})$.*

1. *The ring of regular functions on $\mathcal{V}$ is $\mathbb{C}[q, A, B, C]$; in particular its units are the nonzero constants.*
2. *$\mathcal{V}$ is not affine.*
3. *In the classical topology, $\mathcal{V}$ is simply connected.*

Proof. (1) Write $S = \mathbb{C}[q, A, B, C]$. The open sets $\{A \neq 0\}$, $\{B \neq 0\}$, $\{C \neq 0\}$ cover $\mathcal{V}$ and have rings of regular functions S_A, S_B, S_C, so the ring of regular functions on $\mathcal{V}$ is $S_A \cap S_B \cap S_C$ inside the fraction field of S. If $u = a/A^i = b/B^j$ with $a, b \in S$, then $aB^j = bA^i$; since S is a unique factorization domain and A, B are non-associate primes, A^i divides a, so $u \in S$.

(2) If $\mathcal{V}$ were affine, the inclusion $\mathcal{V} \hookrightarrow \mathbb{A}^{m+3}$ would be a morphism of affine varieties inducing an isomorphism of coordinate rings by (1), hence an isomorphism of varieties [Har77, Corollary I.3.7]. It is not surjective, a contradiction. (Compare [Har77, Exercise I.3.6].)

(3) $\mathbb{C}^3 \setminus \{0\}$ deformation retracts onto the sphere S^5, so $\mathcal{V}$ is homotopy equivalent to S^5, which is simply connected [Hat02, Proposition 1.14]. □

Theorem 7.5. *The space $\mathcal{R}_\Gamma$ is isomorphic to $\mathbb{A}^{57}$, so Question 1.2 applies with $g = 4$ and $N = 57$; for $d = 4$ it proposes $\mathbb{C}^* \times \mathbb{C}^{56}$. However, $\mathcal{R}_\Gamma^{(4)} \cong \mathbb{A}^{54} \times (\mathbb{A}^3 \setminus \{0\})$ is not affine, has only constant units, and is simply connected. Therefore $\mathcal{R}_\Gamma^{(4)}$ is not isomorphic to $\mathbb{C}^* \times \mathbb{C}^{56}$ as an algebraic variety, and not homeomorphic to it in the classical topology.*

Proof. The description of $\mathcal{R}_\Gamma^{(4)}$ is Corollary 5.4, and its properties are Lemma 7.4. The variety $\mathbb{C}^* \times \mathbb{C}^{56}$ is affine, its coordinate ring contains the nonconstant unit given by the coordinate of $\mathbb{C}^*$, and its fundamental group is $\pi_1(S^1) \cong \mathbb{Z}$ [Hat02, Theorem 1.7], since it deformation retracts onto a circle. □

Theorem 7.5 proves Corollary 1.4(2).

Remark 7.6. For the other values of d the prediction of Question 1.2 is correct for this semigroup. For $d = 3$ it gives $(\mathbb{C}^*)^2 \times \mathbb{C}^{55}$, which is isomorphic to $\mathcal{R}_\Gamma^{(3)}$ by Corollary 5.4, and for $d = 5 = g + 1$ it gives $\mathbb{C}^{57} = \mathcal{R}_\Gamma$. Only $\mathcal{R}_\Gamma^{(4)}$, whose complement $\mathcal{E}_5$ has codimension three, fails to have the predicted form.

Appendix A. Expanded equations

Using (4.2), (4.4), (4.5) and (4.6), the equations $r_{47}, r_{48}, r_{49}, q_{59}$ of Theorem 4.5 are

$$r_n = R_n - A\,w_n \quad (n = 47, 48, 49),$$

where

$$\begin{aligned}
R_{47} = {} & -z_{37} - x_{13} + 4h_{14} + 8h_{12}h_{13} + 4h_{12}^3 - 4h_{12}^2 z_{35} + h_{12}z_{35}^2 - 4h_{12}z_{36} \\
& + h_{13}z_{35} - x_{12}z_{35} + z_{35}z_{36}, \\
R_{48} = {} & -z_{38} - x_{14} + 4h_{15} + 8h_{12}h_{14} + 6h_{13}^2 + 12h_{12}^2 h_{13} + h_{12}^4 - 4h_{12}h_{13}z_{35} \\
& + h_{13}z_{35}^2 + h_{14}z_{35} - 4h_{12}^2 z_{36} + h_{12}z_{35}z_{36} - 4h_{12}z_{37} + z_{35}z_{37} \\
& - x_{12}z_{36} - x_{13}z_{35}, \\
R_{49} = {} & -z_{39} - x_{15} + 4h_{16} + 8h_{12}h_{15} + 12h_{13}h_{14} + 12h_{12}^2 h_{14} + 12h_{12}h_{13}^2 + 4h_{12}^3 h_{13} \\
& - 4h_{12}h_{14}z_{35} + h_{14}z_{35}^2 + h_{15}z_{35} - 4h_{12}h_{13}z_{36} + h_{13}z_{35}z_{36} - 4h_{12}^2 z_{37} \\
& + h_{12}z_{35}z_{37} - 4h_{12}z_{38} + z_{35}z_{38} - x_{12}z_{37} - x_{13}z_{36} - x_{14}z_{35},
\end{aligned}$$

and

$$\begin{aligned}
q_{59} = {} & -w_{49} + z_{37} - x_{13} + 2h_{14} + h_{13}z_{35} + h_{13}w_{47} + h_{12}^2 z_{35} - 2h_{12}^2 w_{47} + h_{12}w_{47}^2 \\
& - h_{12}w_{47}z_{35} + 2h_{12}z_{36} - 2h_{12}w_{48} - w_{48}z_{35} + w_{47}w_{48} - w_{47}x_{12} - B\,u_{59}.
\end{aligned}$$

The pivot terms $-z_{37}$, $-z_{38}$, $-w_{49}$, $-z_{39}$ of Theorem 4.5 are visible, and one checks directly that each of these polynomials is homogeneous of the degree given in Proposition 4.4.

Use of AI-assisted tools

AI-assisted tools were used in the research and preparation of this manuscript.

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